Event Cameras for Non-Line-of-Sight Optical Communication: A Survey of Technical Evolution and Future Directions

NLOS-OCC Research Group

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Abstract

Event cameras (neuromorphic vision sensors, dynamic vision sensors) have emerged as a transformative receiver technology for optical camera communication (OCC), fundamentally addressing the temporal sampling bottleneck that limits conventional frame-based cameras to sub-kHz bandwidths. By asynchronously reporting pixel-level log-intensity changes with microsecond resolution and a dynamic range exceeding 120 dB, event cameras enable communication links that are simultaneously high-rate, motion-tolerant, and robust to extreme illumination—properties that are especially valuable when the line-of-sight path is blocked and the receiver must recover data from diffuse reflections. This survey traces the technical evolution of event-camera-based optical communication from its hardware origins (2008–2017) through four developmental stages: proof-of-concept event-OCC (2017–2018), spatial encoding and beacon systems (2022–2024), the first non-line-of-sight (NLOS) event-camera demonstrations (2025), and long-range outdoor plus specialized deployments (2025–2026). We organize the literature around three axes: hardware capability, modulation/demodulation design, and channel environment (line-of-sight vs. NLOS vs. underwater). For each stage we distil the key technical mechanisms—OOK, N-pulse modulation, event interval modulation, toggle demodulation, 8b/10b clock recovery, dynamic QR-code spatial encoding, and adaptive reflection-based NLOS schemes—and tabulate the experimentally demonstrated data-rate-vs-distance frontier. We close with open challenges and a roadmap toward event-camera-enabled NLOS communication networks for 6G-era applications.

Index Terms—Event camera, neuromorphic vision sensor, dynamic vision sensor (DVS), optical camera communication (OCC), non-line-of-sight (NLOS), visible light communication (VLC), rolling shutter, modulation, clock recovery, 6G.

1 Introduction

Optical Camera Communication (OCC) uses an image sensor as the receiver for visible light communication (VLC), exploiting the camera's spatial resolution to discriminate among multiple LED transmitters and reject ambient interference. The fundamental limitation of conventional frame-based OCC is temporal: a commodity camera captures 30–60 frames per second, so the effective sampling bandwidth is below 1 kHz, even with rolling-shutter stripe extraction. LED transmitters, by contrast, are modulated in the MHz range. This sampling mismatch—three to four orders of magnitude—caps the throughput of frame-based OCC and makes it acutely sensitive to motion blur and frame-synchronization errors.

Event cameras, also called Dynamic Vision Sensors (DVS) or neuromorphic vision sensors, dissolve this bottleneck. Each pixel operates asynchronously and independently, generating an *event* (x, y, t, p) only when the log-intensity change exceeds a threshold, where t is a microsecond-resolution timestamp and $p \in \{+1, -1\}$ is the polarity (brightening vs. dimming). The result is a sparse, asynchronous stream with three properties that are ideal for OCC: (i) **high temporal resolution** ($\sim$1 μs, equivalent to $\sim$20–1000 kHz effective sampling), (ii) **wide dynamic range** (>120 dB, vs. $\sim$60–70 dB for CMOS cameras), and (iii) **low data volume and latency** (only changing pixels are transmitted, eliminating redundant background frames).

These properties become even more critical in Non-Line-of-Sight (NLOS) OCC, where the direct LED-to-camera path is blocked and the receiver must recover data from diffuse reflections off walls, the ground, or ordinary objects. The reflected signal is drastically attenuated, smeared by the bidirectional reflectance distribution function (BRDF) of the surface, and superimposed on heterogeneous background illumination. An event camera's high dynamic range can capture the faint reflected intensity changes that a frame-based camera would bury in quantization noise, and its asynchronous operation eliminates the frame-sync problem that plagues reflected signals with unknown propagation delay.

Table 1: Experimentally demonstrated event-OCC performance frontier.

Year	Stage	Data Rate	Distance	BER	Key Technique
2017	**Stage 1**	500 bit/s	5 m	$< 10^{-4}$	OOK + DVS128 receiver [10]
2018	**Stage 1**	16 kbit/s	10 m	N/A	i-PPM + FSK multi-level [11]
2022	**Stage 2**	500 bit/s	100 m	0 (error-free)	Smart visual beacon + LED [17]
2024	**Stage 2**	2.5 kbit/s	0.5 m	94% acc.	Dynamic environment framework [16]
2024	**Stage 2**	—	—	—	N-pulse modulation [1]
2025	**Stage 3**	114 kbit/s	1.25 m	1%	Dynamic QR spatial encoding [13]
2025	**Stage 3**	—	indoor	—	First NLOS event-camera OCC [7]
2025	**Stage 4**	60 kbit/s	200 m	$< 10^{-3}$	OOK + toggle + digital PLL [15]
2025	**Stage 4**	30 kbit/s	400 m	$< 10^{-3}$	(outdoor, same system) [15]
2025	**Stage 4**	28 kbit/s	10 m	—	Event Interval Modulation (EIM) [14]
2026	**Stage 4**	80 kbit/s	320 m	$< 10^{-3}$	8b/10b + OOK-NRZ + PLL, 120° FoV [9]
2026	**Stage 4**	240 kbit/s	320 m	$< 10^{-3}$	Multi-node (3 beams, same system) [9]

Scope and contributions. This survey focuses specifically on event cameras as receivers for optical communication, with emphasis on the progression toward NLOS operation. We trace the technical evolution through four developmental stages (Section 3), analyze the modulation and demodulation design space (Section 4), examine the NLOS event-camera channel in detail (Section 5), and close with open challenges and a roadmap (Section 6). We summarize the experimentally demonstrated data-rate-vs-distance frontier in Table 1 and identify the key technical transitions that enabled each advance.

2 Hardware Foundations

The event-camera hardware evolution directly enabled the communication performance gains surveyed in this paper. Three hardware milestones are foundational:

DVS128 (2008). The first commercial event camera, introduced by Lichtsteiner *et al.* [6], featured a 128×128 pixel array with $40\,\mu$m pixel pitch, 120 dB dynamic range, and $15\,\mu$s latency. Its 350 nm CMOS process produced high dark current, but the asynchronous pixel architecture—a continuous-time logarithmic photoreceptor coupled to a self-timed threshold comparator—established the event-generation model that all subsequent sensors inherit. The DVS128's modest resolution and high noise limited early event-OCC experiments to short ranges and low data rates.

DAVIS240 (2014). Brandli *et al.* [3] introduced the DAVIS240, a 240×180 sensor with 130 dB dynamic range and $3\,\mu$s latency. Crucially, DAVIS240 interleaves event data with global-shutter intensity frames, enabling hybrid processing pipelines that use frames for transmitter localization and events for high-speed demodulation.

Samsung VGA DVS (2017). Son *et al.* [12] demonstrated a 640×480 VGA-resolution DVS with $9\,\mu$m pixels and 300 Meps (million events per second) throughput, fabricated in a BSI CIS process. The higher spatial resolution and event throughput directly enabled longer-range event-OCC links, as the receiver could resolve smaller LED images at greater distances while maintaining sufficient event density for demodulation. Subsequent sensors (e.g., Prophesee EVK4 with Sony IMX636, 1280×720, $4.86\,\mu$m pixel) continued this trend, reaching the $\sim$1 MHz per-pixel event-rate ceiling exploited by the latest long-range demonstrations [9].

3 Technical Evolution: Four Stages

The application of event cameras to optical communication has progressed through four identifiable stages, each defined by a key technical transition. Table 1 summarizes the experimentally demonstrated performance milestones.

3.1 Stage 1: Proof-of-Concept Event-OCC (2017–2018)

The first demonstrations that an event camera could serve as an OCC receiver appeared in 2017–2018. Perez-Ramirez *et al.* [10] demonstrated a 500 bit/s link over 5 m with BER $< 10^{-4}$ using On-Off Keying (OOK)

and a DVS128 receiver. The key insight was that an OOK-modulated LED produces a rising edge (ON event) on the pixels imaging the LED when it turns on and a falling edge (OFF event) when it turns off; the event stream thus directly encodes the bit sequence without the rolling-shutter stripe extraction required by frame-based cameras.

Shen *et al.* [11] extended this to multi-level modulation, achieving 16 kbit/s at 10 m using inverse Pulse Position Modulation (i-PPM) combined with Frequency Shift Keying (FSK). This work first exploited the event camera's microsecond temporal resolution for frequency-domain encoding, moving beyond binary OOK. However, BER was not evaluated, and the system required a static, aligned link—no motion or NLOS capability was demonstrated.

Stage 1 summary. These works established that event cameras can demodulate optical signals at rates far exceeding the frame-rate limit, but the demonstrated throughput ($\leq$16 kbit/s) and range ($\leq$10 m) remained modest, and all links were line-of-sight and static.

3.2 Stage 2: Spatial Encoding and Beacon Systems (2022–2024)

The second stage introduced spatial encoding—using the event camera's 2D pixel array as a communication resource, not just a single-pixel detector—and extended the link range by an order of magnitude.

Wang *et al.* [17] proposed a "smart visual beacon" architecture at IROS 2022, combining an LED transmitter with event-camera demodulation algorithms. They achieved 100 m at 500 bit/s with zero reported bit errors, quantitatively evaluating the tradeoff among LED transmission rate, communication distance, and message accuracy. The beacon architecture was designed for robotic applications—the event camera on a robot demodulates a beacon on a stationary fixture—and it demonstrated that event-OCC could operate at ranges relevant to indoor navigation and V2X scenarios.

Aranda *et al.* [1] introduced N-pulse modulation, a scheme specifically designed for the event camera's operating principles. Rather than encoding each bit as a single intensity transition, N-pulse encodes a symbol as a group of N pulses with a defined pattern, adapting the encoding to the packet's bit composition. This significantly reduced demodulation complexity compared to OOK and PPM, as the receiver could exploit the temporal structure of the pulse group rather than relying on per-edge thresholding. The same group subsequently evaluated pulse-shaped waveform modulation [2], leveraging the sensor's high temporal resolution for shaped pulses that improve noise resilience.

Von Arnim *et al.* [16] addressed the practical challenge of dynamic environments, where ambient light and target motion change during communication. Their framework achieved 2.5 kbit/s with 94% message accuracy at 0.5 m—modest numbers, but the contribution was robustness to changing conditions rather than raw performance.

Stage 2 summary. Stage 2 expanded event-OCC from a single-pixel novelty to a spatially encoded, robot-relevant communication modality. Range extended to 100 m, but throughput remained below 10 kbit/s, and all links were still line-of-sight.

3.3 Stage 3: NLOS Event-Camera Communication (2025)

The third stage—the central focus of this survey—marks the first demonstrations of event cameras for NLOS communication, where the signal is recovered from reflections rather than direct line-of-sight.

3.3.1 First NLOS event-camera OCC

Nishar *et al.* [7] (EWSN 2025, arXiv:2503.11226) demonstrated the first NLOS optical wireless communication using a neuromorphic camera. The key innovation was leveraging *passive reflections from everyday objects* as the communication channel: an LED modulated with OOK or N-pulse modulation illuminates a surface (e.g., a wall, a poster, furniture), and the event camera detects the intensity changes reflected from that surface—even though the camera has no direct view of the LED. The system overcomes the event camera's fundamental limitation (it requires intensity *changes* to generate events; a static scene produces no events) by deliberately inducing VLC-modulated intensity variations, thereby generating events even under zero-motion conditions.

The paper introduced an *adaptive N-pulse modulation* scheme that dynamically adjusts the encoding based on the packet's bit composition, optimizing both data rate and robustness. Experimental results revealed that lighter-colored, glossy objects are better NLOS communication surfaces (higher reflectivity, more specular reflection), while larger objects and matte finishes produce higher error rates due to multipath reflections that smear the temporal structure of the events.

3.3.2 Motion-aware spatial encoding

Su *et al.* [13] (IEEE RAL 2025, arXiv:2412.00816) introduced a complementary advance: a dynamic visual marker (effectively a QR-code-like spatial pattern displayed on a screen) combined with event-based tracking and decoding algorithms. The event camera tracks the marker's corners and decodes the spatially encoded data simultaneously, achieving 114 kbit/s in static conditions—the highest short-range event-OCC throughput to date—and 1 cm localization accuracy with 1% BER under various camera motions. The critical contribution for NLOS is that the event camera's motion-blur immunity enables communication even when the receiver is moving, a scenario that defeats frame-based OCC due to blur-induced stripe distortion. The open-source code and dataset are available at `https://github.com/suhang99/EventOCC`.

Stage 3 summary. Stage 3 established that event cameras can communicate via NLOS reflections and under receiver motion—the two capabilities that frame-based cameras cannot provide. Throughput reached 114 kbit/s (short-range, spatial encoding), but NLOS-through-reflection throughput remained low (order of kbit/s), limited by the reflection channel's attenuation and multipath.

3.4 Stage 4: Long-Range and Specialized Deployments (2025–2026)

The fourth stage, currently unfolding, pushes event-OCC to deployment-relevant ranges and environments: outdoor long-range links, underwater communication, and multi-node networks.

3.4.1 Long-range outdoor

Sumino *et al.* [15] (arXiv:2510.14266, GLOBECOM 2025) achieved the first sub-10^{-3} BER long-range outdoor event-OCC: 60 kbit/s at 200 m and 30 kbit/s at 400 m, using OOK with toggle demodulation and a digital phase-locked loop (PLL) for clock recovery. The toggle demodulation scheme exploits the event camera's polarity output: each OOK symbol produces an ON-event burst followed by an OFF-event burst, and the demodulator toggles its bit decision based on the event polarity, providing intrinsic noise rejection. The digital PLL tracks the symbol clock from the event timestamps, handling the timing jitter that outdoor turbulence introduces.

Nardo *et al.* [9] (arXiv:2609.29776) pushed this further: 80 kbit/s at 320 m with a 120° field-of-view fisheye lens, using 8b/10b line coding with OOK-NRZ and a Bang-Bang PLL. The 8b/10b coding guarantees at least one edge transition every five symbols, ensuring continuous event generation even for long runs of identical bits. In multi-node operation, three parallel beams yielded 240 kbit/s aggregate, with a real-time processing latency under 10 ms per 50 ms event batch. The system used a Prophesee EVK4 (Sony IMX636) sensor; the authors identified the sensor's readout bottleneck (∼100 lines/μs) as the factor preventing the theoretical 1 MHz bandwidth from being reached in practice.

3.4.2 Novel modulation: Event Interval Modulation

Sumino *et al.* [14] (arXiv:2510.14245) proposed Event Interval Modulation (EIM), the first modulation scheme designed specifically for event cameras. Rather than encoding information in the presence/absence of events (OOK) or in pulse patterns (N-pulse), EIM encodes information in the *time intervals between consecutive events*. This fully exploits the event camera's asynchronous temporal resolution and enables higher-order modulation: the interval length maps to a symbol, so M-ary EIM carries $\log_2 M$ bits per event. The proof-of-concept achieved 28 kbit/s at 10 m and 8.4 kbit/s at 50 m indoors, setting a new benchmark for event-OCC bit rate in the moderate-range regime.

3.4.3 Underwater event-camera ISAC

Fang *et al.* [4] (Frontiers in Marine Science 2026) extended event-camera OCC to the underwater domain, demonstrating an event-based camera-enabled underwater optical integrated sensing and communication (ISAC) system. This is notable as the first cross-medium event-OCC: the underwater channel's wavelength-dependent absorption and particulate scattering are analogous to an extreme NLOS channel, and the event camera's high dynamic range is essential for coping with the rapid intensity fluctuations caused by water turbulence.

3.4.4 MISO and real-time systems

Muñoz *et al.* [8] (IEEE PTL 2026) designed a feedback-aware MISO event-OCC system with an LED array transmitter and event camera receiver, achieving near-real-time operation by leveraging event-driven sensing

and a lightweight demodulation scheme. The feedback channel allows the transmitter to adapt its modulation parameters based on receiver-side event-rate feedback, closing the loop for adaptive event-OCC.

Stage 4 summary. Stage 4 has brought event-OCC to deployment-relevant regimes: 320 m outdoor range, 240 kbit/s multi-node throughput, underwater operation, and closed-loop adaptation. The data-rate-vs-distance frontier has advanced by 2–3 orders of magnitude since Stage 1.

4 Modulation and Demodulation Design Space

The event camera's unique sensing model—asynchronous, differential, polarity-labeled—has motivated a distinct modulation design space that differs fundamentally from both photodiode-based VLC and frame-based OCC. We organize the schemes into three categories.

4.1 Edge-Transition-Based Schemes

The simplest and most widely used approach exploits the event camera's native output: each intensity transition generates an event. OOK modulation maps a "1" to an ON-event and a "0" to an OFF-event; the receiver demodulates by checking the polarity of the event stream within each symbol period [9, 10, 15]. The challenge is clock recovery: long runs of identical bits produce no events, causing the receiver to lose symbol alignment. Two solutions have been demonstrated:

- **8b/10b line coding** [9]: guarantees at least one transition every five symbols, ensuring continuous event generation at the cost of 25% overhead.
- **Toggle demodulation** [15]: each symbol period is expected to contain both an ON and an OFF event; the demodulator toggles its decision based on which arrives first, providing intrinsic noise rejection.

4.2 Pulse-Pattern Schemes

N-pulse modulation [1] encodes a symbol as a group of N pulses with a defined temporal pattern, adapting the pattern to the bit composition of the packet. The receiver correlates the event stream against the expected patterns, exploiting the temporal structure for noise rejection. The adaptive variant [7] dynamically selects the pattern based on the packet's bit distribution, optimizing the tradeoff between data rate and robustness. Pulse-shaped waveform modulation [2] further shapes the pulse waveform to exploit the sensor's high temporal resolution.

4.3 Interval-and-Position Schemes

Event Interval Modulation (EIM) [14] encodes information in the time intervals between consecutive events, enabling M-ary modulation where each interval maps to $\log_2 M$ bits. Inverse PPM [11] and FSK [11] encode information in the temporal position and frequency of event bursts, respectively. These schemes fully exploit the microsecond temporal resolution but require precise timing extraction and are sensitive to event jitter.

4.4 Spatial Encoding

The event camera's 2D pixel array enables spatial multiplexing: the dynamic QR-code approach [13] displays a spatially-encoded pattern on a screen, and the event camera decodes the pattern by tracking corner features and reading the spatial cells. This achieves the highest short-range throughput (114 kbit/s) but requires a display transmitter and is limited by the camera's spatial resolution. The smart visual beacon [17] uses a simpler spatial structure (a single LED or small array) with spatial tracking for robotic applications.

5 The NLOS Event-Camera Channel

NLOS event-camera communication is the frontier that most distinguishes this technology from photodiode-based VLC and frame-based OCC. We examine the channel characteristics and the techniques that make NLOS event-OCC feasible.

5.1 The Reflection Channel

In NLOS-OCC, the LED transmitter illuminates a reflecting surface (wall, ground, object), and the event camera images the surface to detect the modulated intensity changes in the reflected light. The channel is characterized by:

- **Path loss**: the reflected optical power is attenuated by the product of the LED-to-surface and surface-to-camera path losses, plus the surface reflectance (typically 0.05–0.9 depending on material and wavelength).
- **BRDF smearing**: the Bidirectional Reflectance Distribution Function of the surface spreads the reflected light over a range of angles, reducing the intensity at the camera and blurring the temporal modulation. Matte surfaces produce diffuse (Lambertian) reflections with severe multipath; glossy surfaces produce more specular reflections that preserve temporal structure.
- **Multipath**: photons reflected via different paths arrive at different times, causing inter-symbol interference. Nishar *et al.* [7] confirmed that larger objects and matte finishes produce higher BER due to multipath, while lighter-colored glossy surfaces are better.

A simplified mathematical model for the NLOS-OCC channel with a patterned reflective surface was developed by He *et al.* [5], providing the channel-characterization foundation for NLOS event-camera work.

5.2 Why Event Cameras Excel at NLOS

Three properties of event cameras are particularly advantageous for NLOS communication:

1. **Dynamic range** (>120 dB): the reflected signal may be 30–60 dB weaker than the direct signal. A frame-based camera with 60–70 dB dynamic range cannot simultaneously capture the bright background and the faint reflected modulation without saturation or quantization loss. An event camera's logarithmic photoreceptor handles this range natively.
2. **Asynchronous operation**: the reflected signal's propagation delay is unknown and varies with geometry. Frame-based OCC requires frame-level synchronization, which fails when the delay is not an integer multiple of the frame period. Event cameras have no frame structure; each event is timestamped independently, so the receiver recovers timing from the event stream itself via a digital PLL [9, 15].
3. **Motion tolerance**: in mobile NLOS scenarios (e.g., a vehicle receiving a reflected signal from a taillight while moving), frame-based cameras suffer motion blur that smears the stripes. Event cameras produce no blur—each pixel reports the intensity change at the moment it occurs, regardless of camera motion [13].

5.3 The Static-Scene Challenge

A unique challenge of event cameras is that a static scene produces no events. In LOS OCC this is not a problem (the LED is modulated, so its pixels always have intensity changes), but in NLOS OCC the reflected intensity change may be too small to exceed the event threshold, especially with matte surfaces. Nishar *et al.* [7] addressed this by deliberately inducing VLC-modulated intensity variations strong enough to generate events via reflection, and by tuning the event threshold to the reflection's contrast level. Future work may exploit adaptive threshold control—dynamically adjusting the event threshold per-pixel based on the local reflected signal strength.

6 Open Challenges and Roadmap

NLOS throughput. The current NLOS event-OCC throughput (order of kbit/s via passive reflection) is 1–2 orders of magnitude below the LOS frontier (80–240 kbit/s). Closing this gap requires: (i) higher-gain optical front-ends (telescopic lenses, Fresnel concentrators) to collect more reflected photons; (ii) adaptive event-threshold control to match the reflection's contrast; and (iii) channel coding (Raptor/RaptorQ fountain codes) tailored to the burst-error profile of the reflection channel.

NLOS channel modeling. Accurate analytical and ray-tracing models for the event-camera NLOS channel—incorporating BRDF, multipath, and the event-generation threshold—are lacking. The simplified model of He *et al.* [5] is a start but does not account for the event camera's differential sensing model. A unified channel model that connects the optical power at the pixel to the event-generation rate is needed.

Modulation for NLOS. The adaptive N-pulse scheme [7] is the only modulation designed for NLOS event-OCC. EIM [14] and 8b/10b+OOK [9] have been demonstrated only in LOS. Extending these to NLOS—where the signal-to-noise ratio is much lower and multipath causes inter-event interference—is an open problem.

Multi-node and networking. The multi-node demonstration of Nardo *et al.* [9] (3 beams, 240 kbit/s) is the only multi-node event-OCC result. A complete networking stack—medium access control, addressing, relaying—for event-camera networks does not yet exist. The IEEE 802.15.7a-2024 standard's AI-PHY and relaying provisions provide a framework, but event-camera-specific adaptations are needed.

Cross-medium and extreme environments. The underwater event-camera ISAC of Fang *et al.* [4] is a promising start, but the underwater channel's absorption and turbulence are far more severe than air. Space-air-ground-sea architectures integrating event-camera OCC with RF and FSO are a long-term vision.

Hardware co-design. The readout bottleneck identified by Nardo *et al.* [9] ($\sim$100 lines/μs) limits the practical bandwidth to well below the sensor's per-pixel event-rate ceiling. Sensor-level innovations—parallel readout, on-chip event processing—are needed to unlock the full $\sim$1 MHz bandwidth.

7 Conclusion

Event cameras have progressed from a vision-sensing novelty to a communication receiver capable of 80 kbit/s at 320 m (LOS outdoor), 114 kbit/s at short range (spatial encoding), and—crucially—kbit/s-rate communication via passive NLOS reflections from everyday objects. The technical evolution has advanced through four stages: proof-of-concept OOK (2017), spatial encoding and beacon systems (2022–2024), the first NLOS event-camera demonstrations (2025), and long-range outdoor plus specialized deployments (2025–2026). The key enablers have been the event camera's microsecond temporal resolution, >120 dB dynamic range, and asynchronous, motion-blur-free operation—properties that are uniquely suited to the NLOS reflection channel. The open challenges are NLOS throughput, channel modeling, NLOS-specific modulation, multi-node networking, and hardware co-design. As event-camera hardware continues to improve and the 6G vision of integrated sensing and communication matures, event-camera-based NLOS OCC is positioned to become a key enabling technology for vehicular, robotic, and IoT networks in cluttered, occluded, and extreme environments.

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